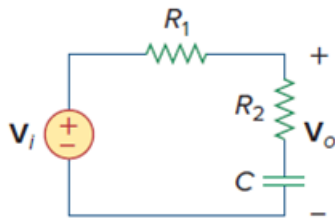


## Problem

Using Fig. 14.69, design a problem to help other students better understand how to determine transfer functions.

**Figure 14.69:**



## Step-by-step solution

## Step 1 of 4

Consider the following circuit diagram:

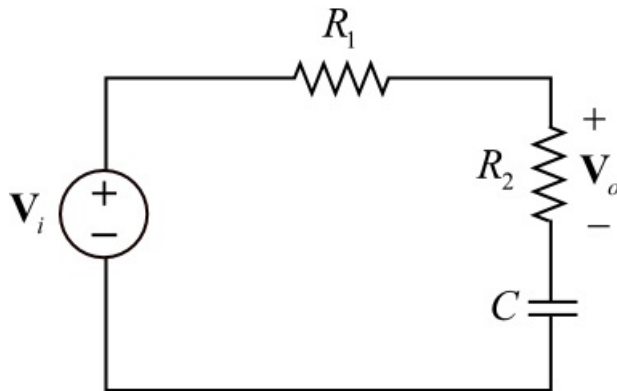


Figure 1

## Step 2 of 4

Let us assume resistance  $R_1 = 5 \, \Omega$

Resistance  $R_2 = 1 \, \Omega$

Capacitor  $C = 1 \, \text{F}$

## Step 3 of 4

We first covert the circuit to the frequency domain:

$$X_c = \frac{1}{j\omega C}$$

$$= \frac{1}{j\omega} \quad \text{since } C = 1 \text{ F}$$

Thus, the frequency domain equivalent circuit is as shown below:

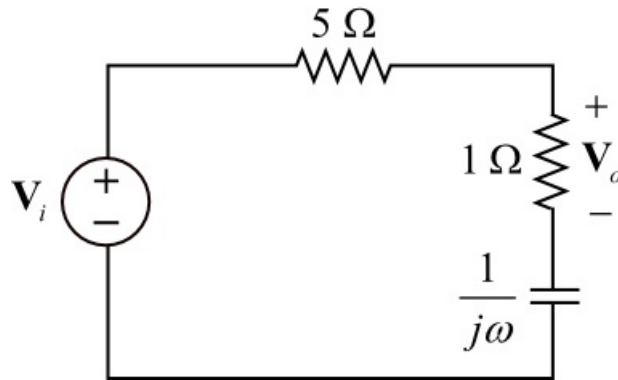


Figure 2

#### Step 4 of 4

Apply voltage division.

$$\begin{aligned} V_o(\omega) &= \left( \frac{1}{5 + 1 + \frac{1}{j\omega}} \right) V_i(\omega) \\ &= \left( \frac{1}{6 + \frac{1}{j\omega}} \right) V_i(\omega) \\ &= \left( \frac{1}{\frac{6j\omega + 1}{j\omega}} \right) V_i(\omega) \\ &= \left( \frac{j\omega}{1 + 6j\omega} \right) V_i(\omega) \end{aligned}$$

Rewrite the expression.

$$\begin{aligned} \frac{V_o(\omega)}{V_i(\omega)} &= \frac{j\omega}{1 + 6j\omega} \\ H(\omega) &= \frac{j\omega}{1 + 6j\omega} \end{aligned}$$

Therefore the transfer function  $H(\omega)$  of the circuit is  $\boxed{\frac{j\omega}{1 + 6j\omega}}$ .